

Foundations of Artificial Intelligence

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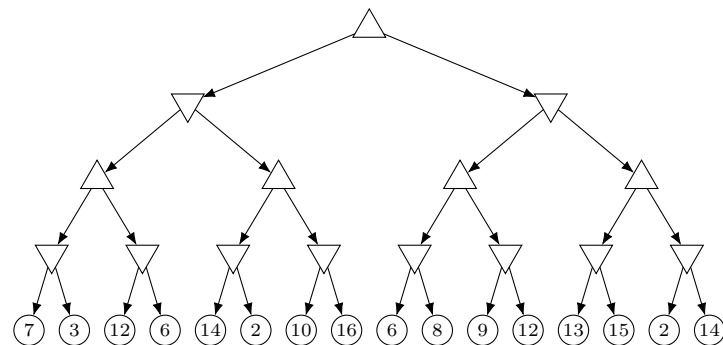
Exercise Sheet 12

Due: May 24, 2026

Please note the submission rules at the end of this exercise sheet. Non-adherence to these rules may lead to your submission not being marked.

Exercise 12.1 (2 marks)

Consider the game inducing the following game tree T . Note that we excluded moves (the edges are unlabeled) because they are irrelevant for the purpose of this exercise.



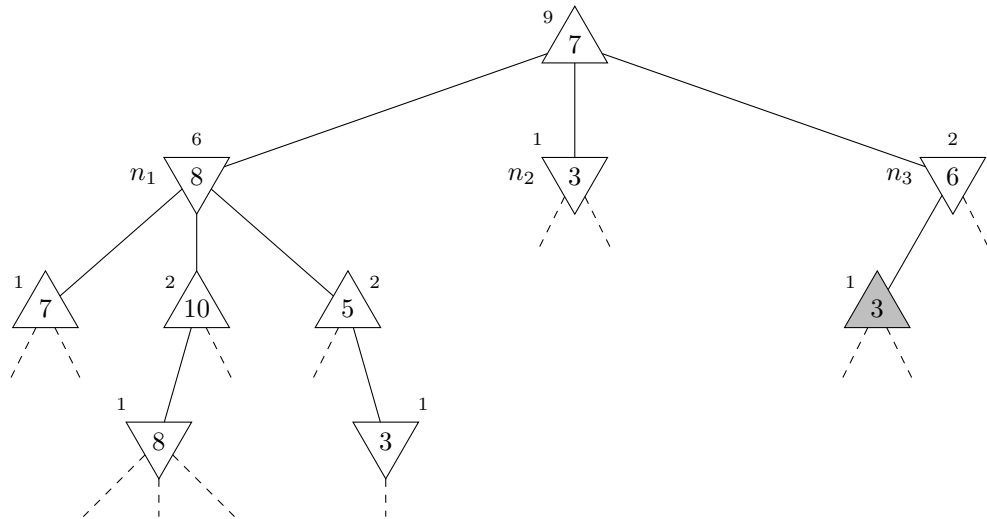
Apply the first 4 iterations of Monte Carlo Tree Search to T with the following policies:

- tree policy: For MAX nodes select the successor with highest utility, for MIN nodes select the successor with lowest utility.
- expansion: Pick the leftmost unexplored child.
- default policy: Pick the rightmost child.

For each iteration, draw the search tree as follows: draw the already explored part of the search tree using solid lines and draw nodes not part of the search tree but encountered during simulation of that iteration using dashed lines. Furthermore, annotate all visited nodes (of the explored part) with the updated utility estimate and visit counter, and annotate the terminal node of the simulation phase with its value.

Exercise 12.2 (0.5+0.5+1+1 marks)

Consider the following partial game tree that was created by Monte-Carlo tree search. State value estimates are given inside each search node and visit counters above each search node. Dashed lines indicate missing successors of search nodes.



- Is there a tree policy that guarantees that Monte-Carlo tree search adds one of the children of the gray search node in the next iteration? Justify your answer.
- Compute the probabilities to choose any of the successors n_1 , n_2 or n_3 of the root node using the ϵ -greedy tree policy with $\epsilon = 0.2$.
- Compute the probabilities to choose any of the successors n_1 , n_2 or n_3 of the root node using the softmax tree policy with $\tau = 10$. Provide how you derived your solution.
- Compute the probabilities to choose any of the successors n_1 , n_2 or n_3 of the root node using the UCB1 tree policy. Provide how you derived your solution.

Submission rules:

- Submit one solution in groups of up to two students, stating the names of all group members in the submission. All group members are equally responsible for all submitted exercises.
- Create a single A4 PDF file for all non-programming exercises. Solutions in $\text{\LaTeX}$ are preferred, but clearly legible scans in A4 with reasonable contrast and file size are accepted as well.
- In case there are programming exercises, upload a ZIP file (ending .zip, .tar.gz or .tgz; *not* .rar or anything else) including your PDF and those code textfiles required by the exercise. Put your names in a comment on top of each code file. Make sure your code compiles and test it. Code that does not compile or which we cannot successfully execute will not be graded.
- Include a statement on scientific integrity at the end of your submission (see below).

Statement on scientific integrity:

- All submitted solutions must be the product of your own work.
- If your written solutions were produced using any help (e.g., from peers or a large language model), list in what way these people or tools contributed to the solutions.
- We do not mark submissions that lack a statement on scientific integrity.
- Examples:

- “We solved all exercises on our own.”
- “We used ChatGPT to explain exercise 7 which we did not understand.”
- “We had no time to solve this exercise sheet and copied all solutions from our friends.”
(If this is the case, thanks for your honesty but this is plagiarism, results in a warning and no marks. It is preferable to submit no solution at all.)